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TECTONIC SIGNIFICANCE OF PALEOMAGNETIC DATA FROM NORTHERN VENEZUELA

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Abstract. A paleomagnetic study, mostly of basic intrusive rocks of Cretaceous and Tertiary age from northern Venezuela, includes 153 samples from 31 sites, of which only 6 sites failed to give consistent results. Fifteen sites within the Caribbean Mountains have easterly declinations with shallow, mixed positive and negative inclinations. These anomalous vectors are interpreted to be the result of tectonic rotation about vertical axes rather than to reflect anomalous magnetic field orientations associated with a polarity transition. Although these sample sites are within igneous intrusive bodies with no control on the original attitude during acquisition of the natural remanent magnetization, the relative conformity of data from numerous separate intrusions requires an average 90° megatectonic rotation of both the Villa de Cura and Tinaco tectonic belts of the Caribbean Mountains. On the other hand, seven sites from the Paraguana Peninsula (and two sites from the younger Tertiary Falcon basin) have southerly declination and positive inclination. These vectors can be explained by rotation about horizontal east-west axes during Cenozoic folding. A compilation of available paleomagnetic data from rocks of Cretaceous age in the Caribbean indicates that anomalous declinations are widespread in both the Greater Antilles and Caribbean Mountains. We interpret these data to support previous suggestions of megatectonic rotations of a more linear, north-south trending Mesozoic island arc system that included portions of the Greater Antilles and Caribbean Mountains of Venezuela.

Introduction

Paleomagnetic studies of rocks of Mesozoic age from the Caribbean area have frequently demonstrated anomalous equatorial or subequatorial paleomagnetic pole positions [MacDonald and Opdyke, 1972]. These are anomalous in view of abundant data from rocks of both North and South America which establish the existence of a high-latitude paleomagnetic pole for the Mesozoic (see data in the work of McElhinney [1973]).

Watkins and Cambray [1970], Fink and Harrison [1971], Guja and Vincenz [1971], and Vincenz et al. [1973] have interpreted anomalous data in the Greater Antilles as a reflection of perturbations of the earth's magnetic field associated with polarity transition intervals. Steinhauser and Vincenz [1973a] suggested that all recorded equatorial paleopoles, including those of the Caribbean area, were related directly to the magnetic field reversal process. MacDonald and Opdyke [1972], Dasgupta and Vincenz [1975], Vincenz and Dasgupta [1978], and Gose and Schwartz [1977], on the other hand, interpret anomalous paleomagnetic data from the Caribbean area to be the result of tectonic rotation and translation of crustal blocks.

This paper reports both normal and anomalous paleomagnetic declinations from rocks of Cretaceous and Tertiary age from 31 sites in northern Venezuela (Figure 1). We too consider the anomalous data to indicate tectonic rotation of portions of the Caribbean Mountains of Venezuela.

Regional Geologic Setting, Caribbean Mountains of Venezuela

The Caribbean Mountains are a Mesozoic orogenic belt along the southern margin of the Caribbean (Figure 1). The geology of the region has been summarized by Bell [1972], Menendez [1967], Bellizzia [1972a], and Maresch [1974]. Previous studies indicate a general subdivision of the orogen into a series of parallel, roughly E-W trending, fault bounded, structural belts each internally characterized by basic similarities in stratigraphy, metamorphism, and structure. These structural belts are (Figure 1) the following:

Cordillera de la Costa Belt

The northernmost Cordillera de la Costa consists of a sequence of Upper Jurassic-Lower Cretaceous clastic and carbonate sediments metamorphosed to the greenschist and epidote-amphibolite facies (Caracas Group). Precambrian and Paleozoic exposures are presumed to be the basement upon which this transgressive-shelf sequence was deposited. Postmetamorphic calc-alkaline intrusives within this belt have been radiometrically dated at 71-88 m.y. [Martin Bellizzia, 1968]. Santamaria and Schubert [1974] report radiometric age determinations from some of these same intrusives indicating a younger thermal event at 30-35 m.y.

Some metasediments on Margarita Island and other Venezuelan offshore islands may be lateral correlatives of the Caracas Group on the mainland.

Caucagua-El Tinaco Belt

The Caucagua-El Tinaco Belt is characterized by exposures of pre-Mesozoic schists and gneisses. Available radiometric age determinations indicate a Triassic (204-210 m.y.) age for the primary metamorphism and a retrograde metamorphic event reflected by younger dates of 112-117 m.y. [Martin Bellizzia, 1968].

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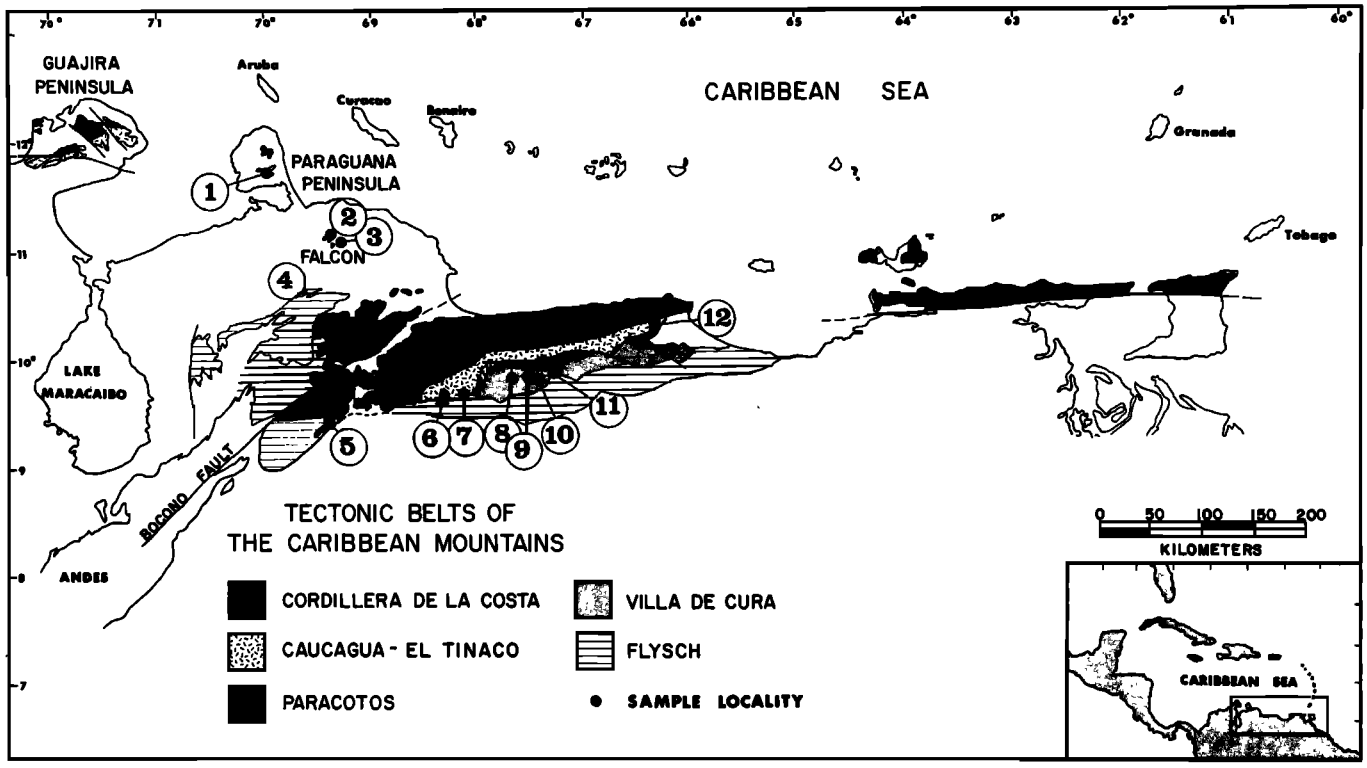


Fig. 1. Generalized geologic map of northern Venezuela showing main sample localities: 1, Santa Ana Complex (PA 1-7, Table 1); 2, Cerro Cienaga; 3, Cerro Redondo; 4, Siquisique; 4, Cerro Pelon (CP 1-3); 6, Rio Tinaco (TC-1); 7, Quebrada Los Bagres (TC-2); 8, Chaparrote; 9, Copei; 10, San Gregorio; 11), El Chacao; 12, Quebrada Catsupo.

Menendez [1962] and Seiders [1962] have reported exposures of Caracas Group and Younger metasediments as well as unmetamorphosed Cretaceous foreland sediments within this pre-Mesozoic tectonic belt. The interpretation of the contact relationships of these Cretaceous metasediments and sediments with the older, underlying schists and gneisses of the Tinaco belt, as well as correlation of the Cretaceous stratigraphy with that in the Cordillera de la Costa tectonic belt, is critical to understanding the Tinaco Belt. All previous workers have interpreted the belt as a rigid 'basement' massif separating the more highly metamorphosed sediments of the Cordillera de la Costa from the less metamorphosed and unmetamorphosed sediments to the south [Menendez, 1962]. In this paper, however, we conclude that the Tinaco belt is allochthonous.

Paracotos Belt

The Paracotos Belt is a narrow, fault bounded zone of predominantly Upper Cretaceous, incipiently metamorphosed sediments of the Paracotos Formation. Serpentinite and gabbro masses are abundant within the fault zones.

Villa de Cura Belt

The Villa de Cura Belt consists of a 4 to 5-km-thick sequence of metavolcanic and metavolcaniclastic units. The belt is approximately 250 km long and bounded by the Agua Fria Fault of the Paracotos Belt to the north and by thrust faults along the southern margin. The volcanic sequence has been metamorphosed under high-pressure/low-temperature conditions [Piburn, 1967]. One available K-Ar radiometric age determination from an overlying volcanic sequence yielded an age of 100 m.y. [Piburn, 1967] and gives a minimum age for the volcanism and metamorphism.

While a few workers [Oxburgh, 1966; Murray, 1972a; Harvey, 1971] have considered the Villa de Cura Belt to be pre-Cretaceous basement, most have interpreted the belt to be an allochthonous mass derived from the north in Late Cretaceous-Paleocene time [e.g., Menendez, 1962, 1967; Bellizzia, 1972a, b; Bell, 1972]. There is less of a consensus, however, regarding the specific position of the Villa de Cura Belt in Mesozoic reconstructions.

Bell [1971] and Piburn [1967] believed that the Villa de Cura Group originally overlay the Caracas Group metasediments of the Cordillera de la Costa Belt. Bellizzia [1977a, b], Maresch [1974], and Skerlec [1974] interpreted the Villa de Cura Belt as the remnant of a bordering orogenic sequence, a fragment of oceanic crust or island arc that was once located north of the present continental margin.

Support for derivation of the Villa de Cura Belt from north of the present continental margin rather than from above the Cordillera de la Costa Belt is based on stratigraphic studies [Oxburgh, 1966; Skerlec, 1979] that establish an unbroken Upper Jurassic through Maestrichtian sedimentary

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sequence within the Cordillera de la Costa Belt. The 4-km-thick metavolcanic sequence of the Villa de Cura Group was not derived from this stratigraphic sequence; nor is there evidence of the abrupt facies changes that would be required if the Villa de Cura Belt is autochthonous and correlative with the adjacent stratigraphic sequences.

The volcanic rocks of the Villa de Cura Formation are described as spilites and keratophyres [Shagam, 1960; Piburn, 1967]. Murray [1972a, b] has noted their similarity to the island arc tholeiite series of Jakes and Gill [1970], as well as the andesitic affinities of rocks in the Tiara Formation that conformably overlies the Villa de Cura Group; he suggested that the Tiara volcanic rocks were an extrusive suite genetically related to zoned ultramafic complexes within the Villa de Cura Belt. These ultramafic rocks were interpreted as deep, subvolcanic cumulates below an andesitic volcanic arc.

Whereas the Villa de Cura Formation contains glaucophane and lawsonite bearing mineral assemblages reflecting high-pressure, low-temperature metamorphism, the ultramafic rocks are completely unmetamorphosed. The anomalous paleomagnetic data reported in this study are primarily from sites in these postmetamorphic intrusives. We interpret these anomalous data as resulting from tectonic rotation, and therefore the metamorphism of the Villa de Cura Formation and intrusion of zoned ultramafic masses occurred prior to the postulated rotation. The Villa de Cura source area is here interpreted as an arc/trench zone active in the early Mesozoic. The islands north of the Venezuelan coast may represent the roots of this island arc complex [Maresch, 1974].

Flysch Belt

Shales, graywackes, conglomerates, slump deposits, and wildflysch of Late Cretaceous-Early Tertiary age form a narrow, intensely deformed structural belt along the southern (Guarico Flysch Basin) and western (Barquisimeto Trough) margins of the Caribbean Mountains. This belt separates the metamorphosed sediments and volcanic rocks of the Caribbean Mountains from the coeval, unmetamorphosed sediments of the foreland. The flysch belt contains material derived from all of the structural belts of the Caribbean Mountains as well as thrust sheets of the foreland shelf sequence.

Numerous exotic, allochthonous masses of gabbro, volcanic, ultramafic intrusive rocks and pelagic cherts petrologically similar to those of the Villa de Cura Belt, but too small to be shown in Figure 1, occur within the flysch belt. These include the Cerro Pelon Ultramafic Complex, Fila de Moroturo, and Cerro Tiramuto [Menendez, 1967; Murray, 1972b; Skerlec, 1974, 1979].

Serpentinite, spilites, pyroxene gabbros, and pelagic cherts also occur in two masses near Siquisique [Compañia Shell de Venezuela, 1965]. These bodies occur along the northern margin of the Barquisimeto Trough and have been interpreted by Bellizzia et al. [1972] as being allochthonous.

Falcon Basin

Between the western termination of the Caribbean Mountains and the Paraguana Peninsula is a large basin of younger, unmetamorphosed Tertiary sediments. A belt of basic igneous rocks is found in the central portion of the Falcon Basin. Gabbros, dolerites, basalts, and andesite porphyries both intrude sediments of Oligocene age and unconformably underlie sediments of the same age [Coronel, 1970; Muessig, 1978]. The Falcon Basin is a distinct younger Tertiary structural province that is separate from the adjacent Mesozoic orogens.

Paraguana and Guajira Peninsulas

A 250-km-wide structural discontinuity separates the western end of the Caribbean Mountains from a similar Mesozoic orogenic belt (Figure 1) exposed on the Paraguana (Venezuela) and Guajira (Colombia) peninsulas [Bellizzia, 1972a, b; Skerlec, 1974]; MacDonald et al. [1971] have correlated the metasediments of the Paraguana Peninsula with those exposed on the Guajira Peninsula and the island of Aruba. Alvarez [1971] has noted similarities between the Colombian Andes and the Mesozoic orogen exposed on the Guajira Peninsula and suggested that they are structurally continuous.

The Santa Ana Igneous Complex [Martin Bellizzia and Arozena, 1972] is exposed in the southern part of the Paraguana Peninsula. Santamaria and Schubert [1974] reported K/Ar whole rock age determinations from gabbros of this complex, of 118, 120 and 129 m.y. (Early Cretaceous). A younger age determination of 63 m.y. was given for an andesite associated with these gabbros. These volcanic, gabbroic, and ultramafic rocks are separated by Cenozoic sediments from a second inlier (Mesa de Crocodite, Figure 4) consisting of a Permian granodiorite overlain by Late Jurassic metasediments [MacDonald, 1968].

Paleomagnetic Sampling

Oriented samples were collected from 31 sites at 12 general localities (Table 1 and Figure 1). Cores 2.5 cm in diameter were either collected in the field with a small, portable, gasoline-powered drill [Doell and Cox, 1967] or drilled from hand samples on a drill press in the laboratory. Orientation was by magnetic compass together with, whenever possible, either astrocompass [La Rochelle, 1961] or sun compass [Creer and Sanver, 1967]. Core orientation is estimated to be accurate to within 5°.

Most of these oriented samples were collected from two of the geological provinces in northern Venezuela: the Caribbean Mountains and Paraguana Peninsula; two sites were located in the Falcon Basin (Figure 1).

Caribbean Mountains

Metavolcanic and metasedimentary rocks, as well as granitic and dioritic rocks from 6 sites, (25 samples) in the Caribbean Mountains yielded weak and inconsistent results (see Table 1) and are not shown in Figure 1. The 79 samples from 16 sites which gave consistent paleomagnetic data were obtained from the Villa de Cura Belt (localities 8, 9, 10, and 11), the Tinaco Belt (localities 6, 7, and 12), and the Flysch Belt (localities 4 and 5). All of these samples were

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TABLE 1. Paleomagnetic Data

Locality	Sites	N	D_n	I_n	k	J_n	χ	OFD	D	I	k	α95
Paraguna												
1	PA1	4	298	38	3.4	4^{-2}_{-6}	3^{-4}_{-5}	500	209	42	4.5	49.2
	PA2	4	146	58	6.6	5^{-4}	2^{-4}	200	163	54	14.4	20.8
	PA3	4	165	-26	3.1	3^{-5}	2^{-5}	500	174	31	15.5	20.0
	PA4	5	211	64	1.8	5^{-5}	7^{-5}	300	216	26	12.7	19.5
	PA5	4	114	3	1.8	1^{-5}	3^{-5}	300	141	21	30.3	11.1
	PA6	6	194	31	72	5^{-5}	3^{-5}	500	192	29	58.1	8.9
	PA7	6	225	21	1.1	6^{-6}	3^{-5}	500	184	24	7.5	26
Falcon												
2	C. Redonda	10	27	58	2.7	3^{-4}	4^{-4}_{-5}	500	194	75	11.6	17.0
3	C. Cienaga	6	353	81	3.1	2^{-6}	4^{-5}	500	193	64	14.2	18.4
Siquisique												
4	Los Algodones	10	323	26	1.9	4^{-6}	2^{-5}	500	235	-14	5.7	25.3
Cerro Pelon												
5	CP1	6	123	40	15.5	1^{-5}_{-4}	4^{-5}_{-5}	200	126	39	12.0	20.1
	CP2	5	138	-1	24.6	5^{-4}	4^{-3}	300	139	-3	27.8	17.7
	CP3	4	70	76	19.0	8^{-4}		500	118	16	16.4	23.4
Chaparrote												
8		8	65	39	3.8	3^{-4}	1^{-3}	500	58	6	15.1	16.0
Copei												
9		5	333	75	40.6	4^{-4}	1^{-3}	500	335	46	5.5	43.2
San Gregorio												
10		7	83	-16	5.7	2^{-5}	$5^{-5}_{3^{-3}}$	300	92	-34	17.1	15.0

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El Chacao												
11	EC1	4	81	34	28.3	2^{-4}	9^{-4}	300	108	18	36.8	15.4
	EC2	4	38	20	9.2	2^{-4}	5^{-3}	200	85	29	35.2	15.7
	EC3	2	62	14	16.2	7^{-4}	3^{-3}	300	79	13	61.3	32.5
	EC4	4	125	84	5.3	7^{-4}	3^{-3}	300	84	-7	7.6	35.8
	EC5	3	94	29	11.1	2^{-3}	9^{-4}	200	107	22	55.4	16.7
	EC6	4	66	47	10.0	7^{-4}	3^{-3}	300	69	12	20.3	20.9
	EC7	4	41	65	17.1	2^{-3}	9^{-3}	500	111	16	54.6	12.5
Tinaco Complex												
6	TC1	4	64	54	18.2	4^{-5}	4^{-5}	200	64	53	16.9	23.0
7	TC2	4	61	18	187.1	2^{-5}		200	66	15	117.9	8.5
Quebrada Catsupo												
12	Conoropa Volcanics	6	84	-63	11.0	7^{-5}	4^{-5}	200	94	-68	21.6	14.7
El Carmen*		5	14	29	4.5	3^{-7}	2^{-5}					
Pilancones*		4	319	18	2.3	1^{-6}	5^{-5}					
Puerto Cabello Eclogite*		2				2^{-7}	6^{-5}					
Guaremal Granite*		4				1^{-7}	2^{-5}					
Todasana Complex*		5	344	81	1.6	5^{-7}	3^{-5}					

Locality numbers refer to those in Figure 1; N is number of specimens, D is declination, I is inclination, k is precision parameter, J_n is intensity of magnetization ($\times 10^{-6}$ emu/cc), χ is specific susceptibility (emu/c.c. · Oe), OFD is optimal demagnetizing field (Oe), $\alpha 95$ is radius of circle of confidence.
*These data were rejected because of lack of within-site consistency or the lack of a stable remanent magnetism.

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collected from basic to ultrabasic, postmetamorphic intrusives of Cretaceous age.

Within the Villa de Cura Tectonic Belt, locality 11 (seven sites) is the El Chacao Ultramafic Complex (Figure 2), a zoned Alaskan-type intrusion described by Murray [1972b]. Localities 9 and 10 (Figure 1) are small consaguineous intrusive bodies, and locality 8 is a swarm of dikes also intrusive into the metavolcanic rocks of the Villa de Cura Group.

Localities 6 and 7 are small ultramafic masses which intrude the gneisses and schists of the Tinaco Complex at the western end of the Tinaco Tectonic Belt [Menendez, 1962]. Locality 12 is a hornblende gabbro intruding the Conoropa volcanic rocks which Seiders [1962] described as conformably overlying the pre-Mesozoic rocks of the Tinaco Tectonic Belt.

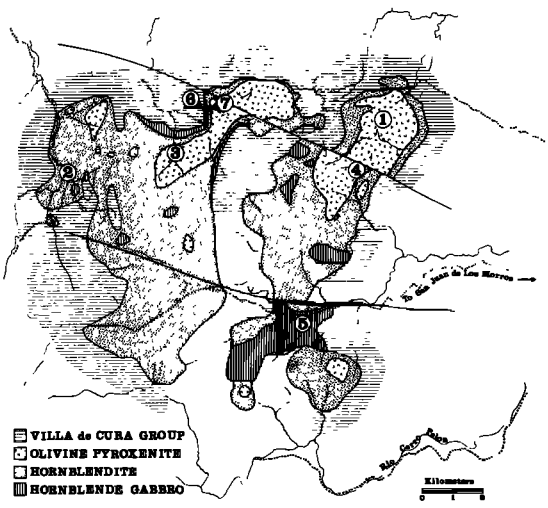
Localities 4 and 5 are within two smaller, allochthonous masses of the Flysch Belt. Locality 5, Cerro Pelon (Figures 1 and 3) is another Alaskan-type zoned ultramafic complex similar to the El Chacao Complex. It also intrudes a metavolcanic sequence similar to the Villa de Cura Group. Locality 4, Siquisique, is in pyroxene gabbros associated with serpentinites, spilites, and pelagic cherts occurring along the northern margin of the Surco de Barquisimeto [Compania Shell de Venezuela, 1965; Bellizzia et al., 1972].

Paraguana Peninsula

A total of 33 samples were collected from seven sites in gabbros and volcanics of the Santa Ana Complex (Figure 4 [after Martin Bellizzia and de Arozena, 1972] in the southern portion of the Paraguana Peninsula, locality 1 (Figure 1).

Falcon Basin

Two sites, consisting of a total of 16 samples, were collected from intrusives of Oligocene age in the Tertiary Falcon Basin (Figure 1,



EL CHACAO ZONED ULTRAMAFIC COMPLEX

Fig. 2. Geologic map of the El Chacao Zoned Ultramafic Complex [after Murray, 1972b] with paleomagnetic sampling sites indicated by numbers.

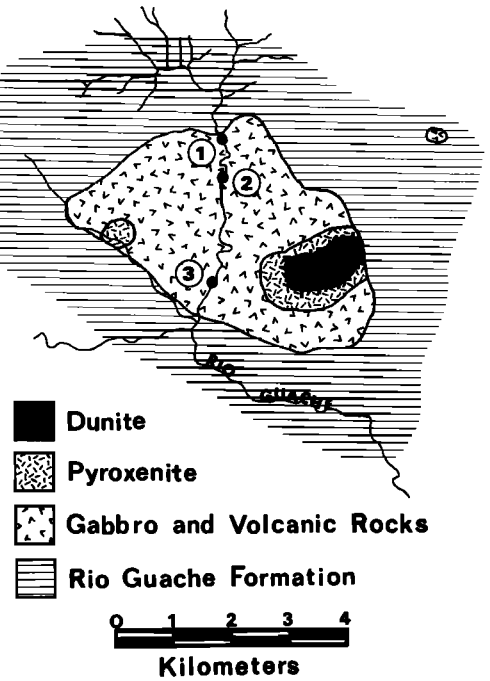


Fig. 3. Geologic map of the Cerro Pelon Ultramafic Complex [after Murray, 1972b; Skerlec, 1979] with paleomagnetic sampling sites indicated by numbers.

localities 2, 3). Samples were collected from massive gabbros and diorites (which, however, have undergone pervasive low-temperature alteration) at Cerro Cienaga (locality 2) and Cerro Redonda (locality 3).

Magnetic Measurements

Natural remanent magnetization (NRM) measurements were made on a Princeton Applied Research, S.M. 1 spinner magnetometer with alterating field (af) demagnetization standard equipment. The results from all samples collected (Figure 1) are listed in Table 1.

At least one specimen from each sample was systematically demagnetized in alternating fields up to 500 Oe (N, 50, 100, 150, 200, 300, 500). A second specimen was then demagnetized at two or three stages only, depending upon the initial results. Site means were calculated for the NRM's and for each af demagnetizing field. That demagnetized vector with the lowest dispersion (k) was selected, and reported, together with the appropriate optimal demagnetizing field (ODF) in Table 1.

Internally consistent sample and site vectors were commonly obtained from specimens with an average Jn between 1 x 10^{-4} and 5 x 10^{-5} emu/cc. Weak samples (Jn < x 10^{-6} emu/cc) invariably gave inconsistent results. Petrographic examination revealed that these weak samples showed prominent evidence of alteration.

With rare exceptions (see Table 1), all samples had specific susceptibility (χ) greater than Jn and hence had Koenigsberger ratios less than 1.

Because most of the rock bodies sampled were intrusive, with no direct evidence as to their

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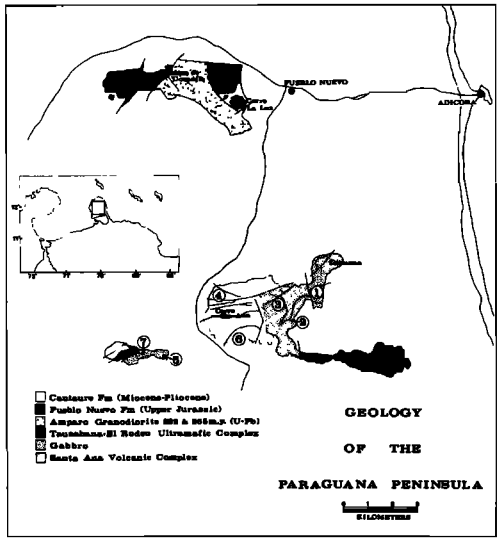


Fig. 4. Geologic map of the Paraguana Peninsula, Venezuela [after Martin Bellizzia and de Arozena, 1972] showing location of paleomagnetic sampling sites.

original attitude at time of magnetization, there was little point in calculating virtual geomagnetic poles; furthermore, most of these rock masses are demonstrably (or inferred to be) allochthonous. Where sampling sites were closely spaced (as at Paraguana, Cerro Pelon, and El Chacao), vector means have been calculated to facilitate graphical presentation.

The strongest samples, giving the most stable and internally consistent vectors, were from sites in basic to ultrabasic rocks belonging to, or consanguineous with, zoned Alaskan-type intrusives. These complexes, varying from pyroxenite through hornblendite to hornblende gabbro, are intrusive into the Villa de Cura and Caucagua-El Tinaco Belts [Murray, 1972b]. Only two demonstrably zoned complexes are known, but smaller hornblende gabbro dikes and sheets are widespread. Examination of such samples in transmitted and reflected light revealed the presence of innumerable fine needles and rodlets of magnetite in all of the ferromagnesian minerals. The origin of this magnetite is inferred to be by exsolution, or high-temperature oxidation of ferrous iron in the original silicate minerals. Magnetite concentrations are commonly associated wtih Alaskan-type intrusions, and the segregation of these and the fine magnetite rodlets is compatible with the inferred petrogenetic history. According to Murray [1972b] this involves crystallization of a hydrous basaltic magma; the high pH_2O is likely to favor a high fO_2, thereby fostering the high-temperature oxidation of ferrous iron. (Note that these data suggest that Alaskan-type ultramafic rocks in the type area (Alaska) are also likely to provide suitable material for paleomagnetic study.)

The results of the paleomagnetic measurements may be generalized as follows:

1. Anorthositic gabbros and volcanic rocks from the Santa Ana Complex, Paraguana Peninsula, have generally south to southwesterly declinations, with positive inclinations (total seven sites, Figures 5 and 6). The younger Falcon Basin plutons also have southwesterly declinations but with very steep positive inclinations (two sites, Figure 6).

2. Diorites, hornblende gabbros, hornblendite, and pyroxenite from basic and ultramafic (Alaskan type) intrusions into the Villa de Cura Group and into gneisses of the El Caucagua-Tinaco tectonic belt have moderately strong, stable, and consistent NRM with variable, easterly declination and shallow, positive, or negative inclination (15 out of 16 sites, Figures 7 and 8).

3. Metavolcanic rocks of the Villa de Cura Group (El Carmen, one site and locality 9 at Copei) and overlying the Tinaco basement (Pilancones, one site), and eclogites, granites, and diorites within and intruding the Caracas group of the Cordillera de la Costa Belt (3 sites) yielded weak and inconsistent results.

Discussion

Two paleomagnetic provinces can be distinguished within the data from northern Venezuela: (1) the Caribbean Mountains, characterized by anomalous easterly declinations and (2) the Paraguana and Falcon basins, characterized by southerly declinations but anomalous positive inclinations.

Falcon Basin and Paraguana Peninsula

Whereas the mean southerly declinations from both the Paraguana Peninsula and the Falcon Basin are not significantly different from those reported from rocks of Cretaceous and Oligocene age in North and South America [Creer, 1970; McElhinny, 1973], the positive inclinations are distinctly anomalous, since this combination would require acquisition of NRM during a period of reversed magnetization at a site in the south-

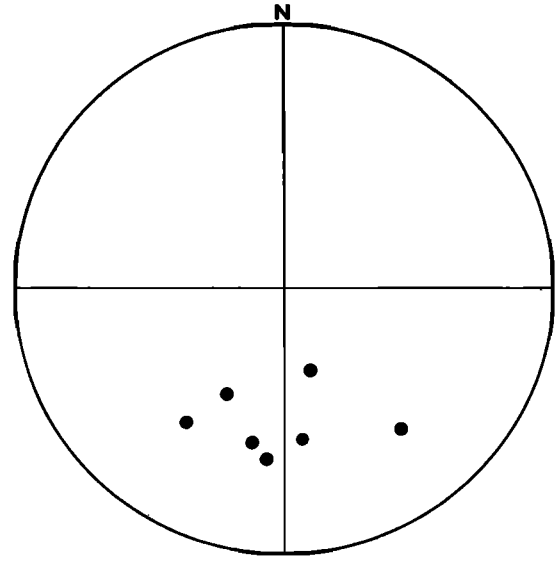


Fig. 5. Mean paleomagnetic vectors of 33 samples from 7 sites in the Santa Ana Complex, Paraguana Peninsula. Data (Figures 5-8) plotted on equal-area projections, lower hemisphere. Solid circles are positive inclinations. Open circles are negative inclinations plotted on the upper hemisphere.

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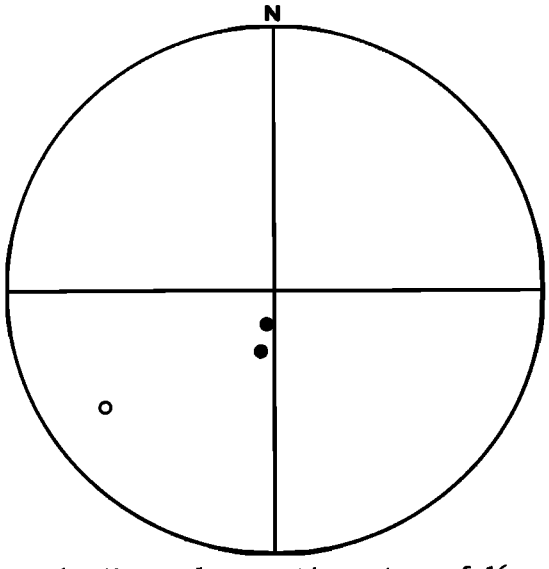


Fig. 6. Mean paleomagnetic vectors of 16 samples from Cerro Cienaga and Cerro Redondo (Falcon Basin) and 10 samples from Siquisique.

ern hemisphere. Data from nuclear South America indicate that these sites have been in the northern hemisphere during the Cretaceous and Cenozoic [Creer, 1970]. Rather than requiring rapid latitudinal shifts of South America it is more reasonable to conclude that these anomalous vectors are the result of rotation associated with regional deformation.

Oligocene sediments of the Falcon Basin, into which the plutons measured were intruded, have been folded about an axis trending N60°E. At least 70° of rotation about this axis would be required to restore these steeply inclined vectors to those expected from a reversed Oligocene magnetic field (Figure 9), but there is no obvious field evidence for rotations of this magnitude in the vicinity of the intrusions sampled.

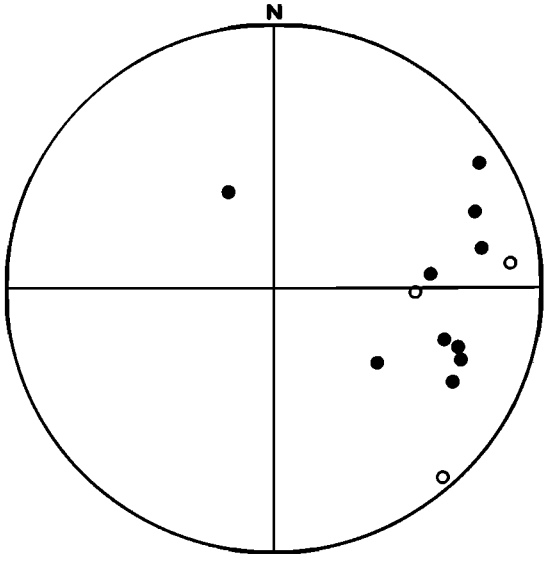


Fig. 7. Mean paleomagnetic vectors of 60 samples from 13 sites within Villa de Cura Tectonic Belt and Cerro Pelon Ultramafic Complex.

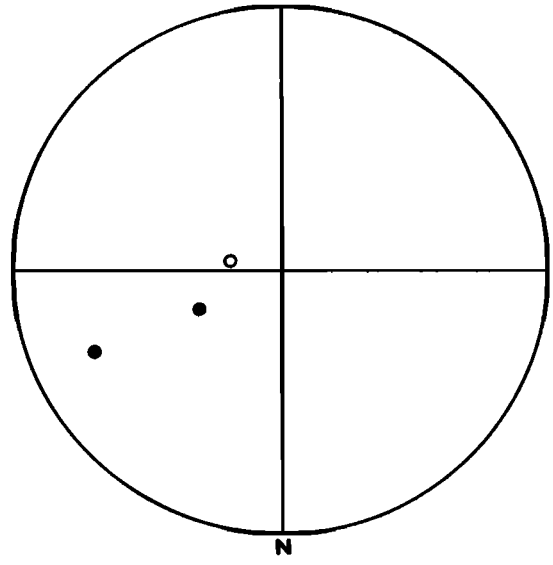


Fig. 8. Mean paleomagnetic vectors of 14 samples from 3 sites in the Tinaco Tectonic Belt.

In view of the intense alteration of these rocks a superimposition of two vectors is suspected: one primary, reversed, prefolding; and a stable secondary, normal, postfolding. This could best explain the high inclination of these sites.

This Cenozoic folding, however, has probably also affected the Santa Ana Complex of the Paraguana Peninsula. Figure 9 shows the rotation path of the measured NRM assuming folding has affected these Lower Cretaceous intrusives. The anomalous inclinations from the Paraguana Peninsula can be easily restored to negative inclinations with southerly declinations by rotations about a horizontal N60°E axis.

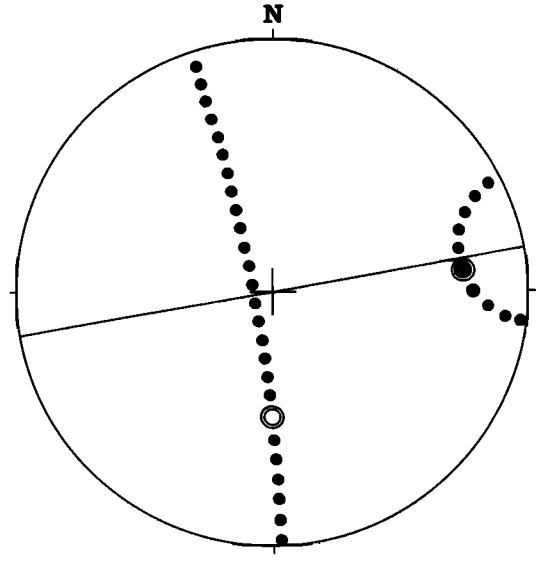


Fig. 9. Rotation paths of mean paleomagnetic vectors about regional fold axis of the Caribbean Mountains. Large open circle is mean of seven sites from the Paraguana Peninsula. Large solid circle is mean of all data from the Villa de Cura Tectonic Belt. (Lower hemisphere, equal-angle projection.)

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The Caribbean Mountains

The anomalous easterly paleomagnetic declinations in the Caribbean Mountain data may be interpreted as resulting from either tectonic rotation or an anomalous magnetic field.

Workers in other areas have suggested that anomalous declinations reflect not tectonic rotations but an anomaly in the earth's magnetic field associated with a polarity transition interval [Goldstein and Strangway, 1969]. We believe this model does not satisfactorily explain the anomalous data from the Caribbean Mountains.

Anomalous declinations are reported from eight separate igneous intrusions distributed over a distance of almost 300 km. It is improbable that all eight, widely separated intrusions acquired their NRM during one polarity transition interval. The duration of a transitional interval (<10,000 years, McElhinny [1973]) is so short in comparison to average periods of normal or reversed polarity that anomalous NRM's resulting from a polarity transition should be less prevalent than meridionally oriented NRM vectors. With the exception of only one site (Copei), meridionally oriented NRM vectors are lacking in our data from the Caribbean Mountains. Furthermore, anomalous declinations are recorded at a large number of sites which individually exhibit a high degree of internal consistency.

We conclude that the observed anomalous easterly declinations must have resulted from rotations of crustal blocks.

Tectonic Rotations in the Caribbean Mountains

There are two possible interpretations of the tectonic rotation, each of which has significantly different implications for the evolution of the southern Caribbean boundary. The first interpretation requires megatectonic rotation of a once north-south trending arc-trench complex about a single rotation axis in northwestern South America. Megatectonic rotations have previously been proposed based on the gross geologic similarity and the sinuous pattern of the Mesozoic orogenic belts of the Caribbean [North, 1965; Case et al., 1971; Roper, 1974].

The second interpretation assumes that the entire Caribbean Mountains formed initially as an east-west trending collision boundary between the Caribbean and South American plates [Bell, 1972]. Tectonic rotation was accomplished by rotation in situ of numerous small crustal blocks. Folding, submarine slumping, thrust, normal, and strike slip faulting have all been involved in the complex structural evolution of the Caribbean Mountains and theoretically could provide a mechanism for rotation of a meridional NRM vector.

There are three reasons for believing that tectonic rotation did not occur by rotation in situ about numerous axes of rotation. First, much of the deformation within the Caribbean Mountains involved rotation about horizontal axes parallel to the east-west trend of the region. Although rotation about a horizontal axis is adequate for explaining the anomalous data of the Paraguana Peninsula, Figure 9 demonstrates the impossibility of similarly explaining the easterly declinations of the Caribbean Mountains. The tectonic rotation in the Caribbean Mountains probably occurred about vertical axes.

Second, at least the 300-km long Villa de Cura Belt underwent tectonic rotation as a structurally coherent mass. Smaller masses such as the Cerro Pelon Complex of Siquisique (Figure 1) may have been derived from a narrow, east-west trending source area as in Figure 10a and possibly have been rotated by thrusting or submarine slumping. Stratigraphic contacts within the Villa de Cura Belt, however, maintain continuity and parallelism along the entire length of the belt [e.g., Piburn, 1967; Oxburgh, 1966]. Minor cross faults noted by Harvey [1971] do suggest a net clockwise rotation of the Villa de Cura Belt by thrusting and strike-slip faulting, but the 10°-15° of rotation thus produced is much less than the 90° of rotation required by the paleomagnetic data.

Third, the intensity of strain within the Villa de Cura Belt is inconsistent with 90° of tectonic rotation by simple shear associated with strike slip faulting. The pattern of strike slip faulting in northern Venezuela [Rod, 1956; Alberding, 1957] is consistent with clockwise tectonic rotation, but it did not produce 90° of rotation in the NRM vectors.

Thus although some small component of the tectonic rotation may have resulted from rotation of crustal blocks in situ, the mechanism does not explain 90° of rotation. The interpretation we favor is that those units that underwent tectonic rotation were once part of a north-south trending arc/trench system that once lay perpendicular to the Venezuelan continental margin and which

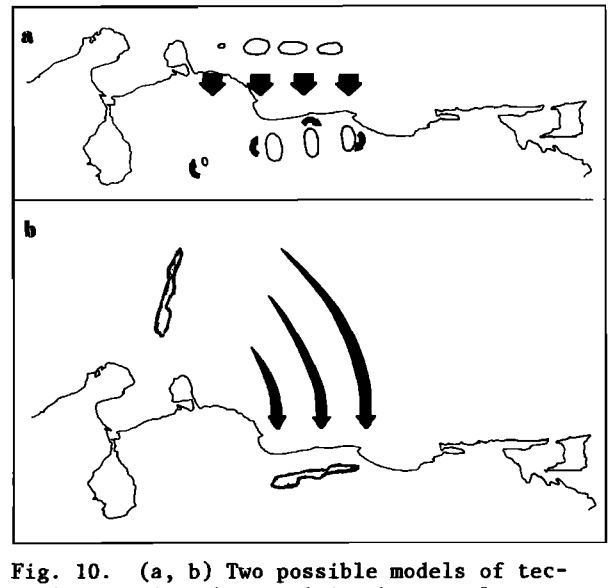


Fig. 10. (a, b) Two possible models of tectonic rotation that explain the anomalous paleomagnetic data. In Figure 10a the Villa de Cura Tectonic Belt and associated rocks were derived from an east-west trending island arc. The rotation was accomplished by submarine slumping or other mechanisms which are discussed in the text. In Figure 10b these rocks were part of a north-south trending island arc system that was originally perpendicular to the Venezuelan continental margin.

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reached its present position by megatectonic rotation.

This same conclusion was reached by several previous workers, but whereas North [1965] and Case et al. [1971] considered the Caribbean Mountains as a single mass that underwent megatectonic rotation, the Caribbean Mountains, in reality, consist of several tectonic belts, each of which has undergone a different tectonic history.

The Caribbean Mountains consist of both an island arc and the stable Venezuelan continental margin with which the arc collided during the Upper Cretaceous [Maresch, 1974; Skerlec, 1974]. Maresch [1974] suggested a collision model in view of the lack of paired metamorphic belts in the Caribbean Mountains. Skerlec [1974] noted the influx of detrital material and allochthonous blocks from an island arc source area during the Upper Cretaceous onto what had previously been a stable, Atlantic type continental margin.

The Tinaco Tectonic Belt and the La Teta Lavas

Not all of the units which underwent tectonic rotations were necessarily part of an arc/trench complex. The La Teta Lavas of the Guajira Peninsula, Colombia, described by MacDonald and Opdyke [1972] apparently underlie Mesozoic platform sediments of nuclear South America. These lavas are not easily correlated with volcanic rocks of island arcs such as the Villa de Cura Group.

Similarly, the gneisses and schists of the Tinaco Belt are not obviously the metamorphosed equivalent of an island arc/trench complex or oceanic crust. The arkosic sediments and the conglomerates intercalated with volcanic rocks of the Tinaco Belt suggest deposition on or marginal to continental crust. The Tinaco Belt has been interpreted by all previous workers [Menendez, 1962] as autochthonous basement upon which the metasediments of the Caribbean Mountains were originally deposited. We do not consider the evidence for an autochthonous Tinaco Belt as conclusive and the relationships noted by Murray [1972a, b] suggest a reinterpretation of the nature of the belt is necessary.

Murray [1972b] first recognized the belt of petrologically similar gabbros and ultramafics within the Caribbean Mountains. It is significant that these rocks intrude both the Tinaco and Villa de Cura tectonic belts. The present spatial relationships of the two tectonic belts indicates that the Villa de Cura Belt overlies the Tinaco Tectonic Belt although separated from it by a narrow belt of Maestrichtian sediments. It is considered unlikely that the Villa de Cura Belt slid southward independently of the Tinaco Belt to a point where the ultramafic and gabbroic intrusives in both tectonic belts now form a single line, all with anomalous easterly paleomagnetic declinations. Rather, both the Villa de Cura and Tinaco tectonic belts necessarily attained their present relative positions prior to intrusion of these rocks and prior to tectonic rotation. Thus if the Villa de Cura Belt is allochthonous, so must be the Tinaco Belt.

The nature of the La Teta Lavas and Tinaco Belt suggest that tectonic rotation may have affected parts of the stable continental margin in addition to the arc/trench complex. The possibility remains that part of the arc/trench complex was built not solely on oceanic crust but upon continental crust or an older orogenic sequence.

Timing of Rotation

Menendez [1967] and Bell [1972] argue on stratigraphic grounds that the Villa de Cura tectonic belt reached its present position in Maestrichtian-Paleocene time. The most important time constraints are that rotation must have occurred after metamorphism of the Villa de Cura Formation and after intrusion of the gabbro-ultramafic suite. The only available radiometric age determination applicable to this problem is the 100 m.y. ± 10 (K/Ar, whole rock) date reported by Piburn [1967] from a metatuff at the base of the Tiara Formation which conformably overlies the Villa de Cura Formation.

Four K/Ar age determinations from gabbro within the Villa de Cura Formation with ages of 65, 66, 67, and 67 m.y. [Santamaria and Schubert, 1974] probably reflect a later thermal event rather than primary crystallization.

Ultramafic and gabbroic material identical to that of the Alaskan-type intrusives appears as detritus for the first time in conglomerates of the Garrapata Formation of Turonian age. These immature clastic sequences may mark the earliest stage of rotation and collision.

Tectonic rotation of the Villa de Cura/Tinaco source area thus apparently occurred after 100 m.y. and was completed by Paleocene time.

Caribbean Paleomagnetic Data

The model of megatectonic rotations is supported by paleomagnetic data compiled from both the northern and southern boundaries of the Caribbean plate.

In Figure 11, data from rocks of Cenozoic age, which include locality 2 in the Falcon Basin as well as localities 9 and 12b on Jamaica, display paleomagnetic declinations which do not differ greatly from those reported from Cenozoic rocks in North and South America [McElhinny, 1973].

The remainder of the data from rocks of Mesozoic age, however, display predominantly anomalous paleomagnetic declinations. It is possibly significant that whereas the majority of the data from the southern margin of the Caribbean are characterized by anomalous easterly declinations, the majority of the data along the northern margin are characterized by anomalous westerly declinations (Figure 11). The reason for this is not known, but if one assumes that both sets of NRM were acquired during one long lasting polarity epoch, then this pattern suggests opposite senses of rotation along the northern and southern Caribbean boundaries. Unfortunately, even if the assumption is correct, the sense of rotation, whether clockwise or counterclockwise, is not unambiguously afforded by the paleomagnetic data alone, since the original NRM may have been either normal or reversed with shallow inclinations.

Regional geology does place constraints on the sense of rotation. Similarities in gross struc-

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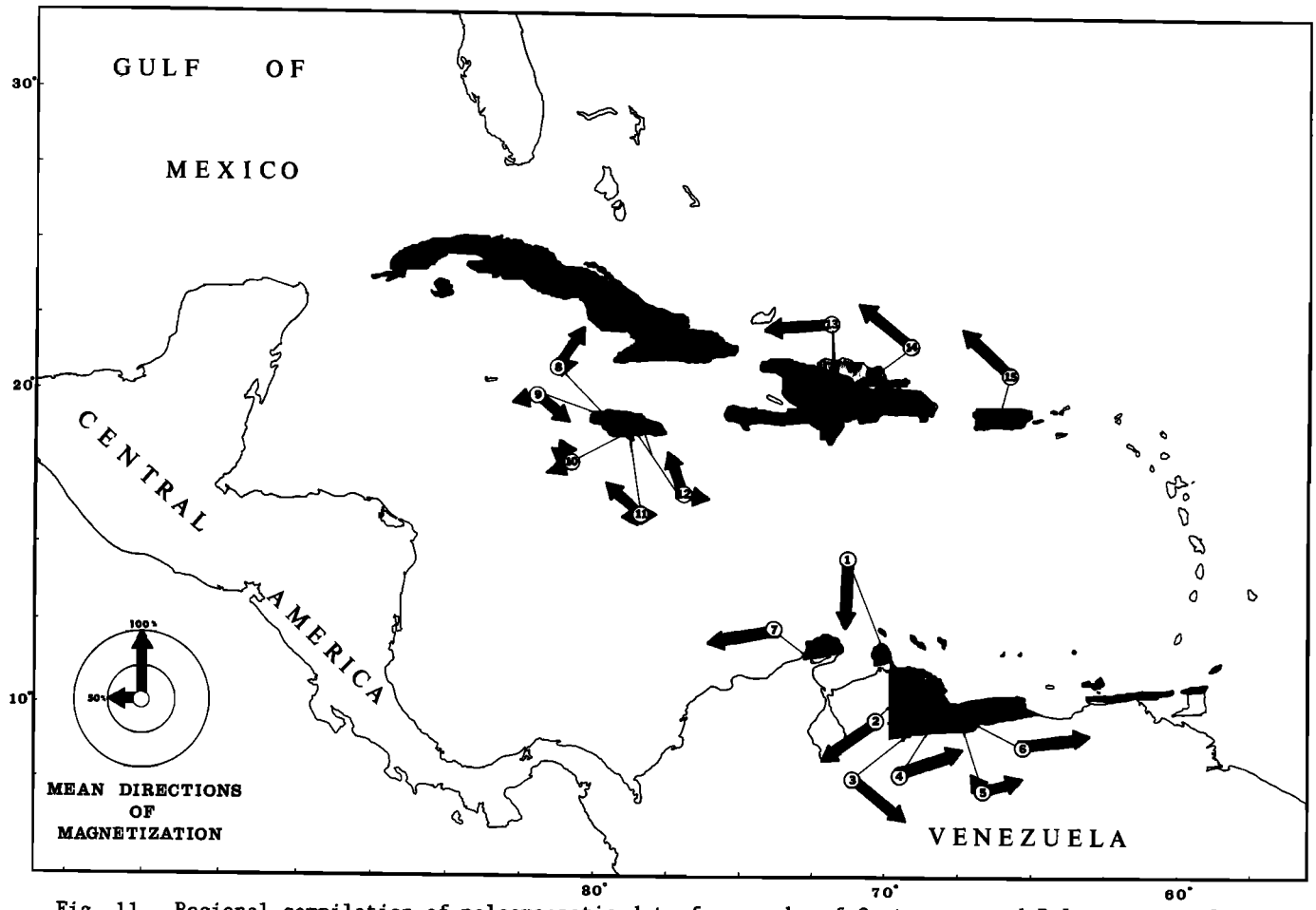


Fig. 11. Regional compilation of paleomagnetic data from rocks of Cretaceous and Paleocene age from the Carribbean. The vectors give the mean declination, and their length is proportional to the percentage of the total samples having that mean direction. For example, inset in lower left shows vectors having 100% and 50% of the samples in the mean direction. Data numbers are 1-6, this study; 7, MacDonald and Opdyke [1972]; 8, Dasgupta and Vincenz [1975]; 9, Vincenz et al. [1973]; 10, Steinhauser and Vincenz, [1973]; 11, Guja and Vincenz [1971]; 12, Watkins and Cambray [1970]; 13, S. A. Vincenz and S. N. Dasgupta (unpublished data, 1979); 14, Vincenz and Dasgupta [1978]; 15, K. L. Fink and C. G. A. Harrison (personal communication, 1978).

ture between the western end of the Caribbean Mountains and the northeastern end of the Colombian orogenic belt [Alvarez, 1971; Skerlec, 1979] suggest the two orogenic belts may have been, in part, structurally continuous or connected by a major trench/trench transform fault [Bell, 1971]. If the prerotation southern terminus of the Villa de Cura arc/trench system originally lay in the vicinity of the Guajira Peninsula, the Villa de Cura Belt has rotated clockwise rather than counterclockwise. If the anomalous easterly declinations of the Caribbean Mountains reflect clockwise rotation, then on the assumption made above, the anomalous westerly declinations of the Greater Antilles reflect counterclockwise rotations.

The reconstruction of an Early Mesozoic arc/trench complex consisting of the Greater Antilles and portions of the Caribbean Mountains places critical constraints on the origin of the Caribbean. This arc/trench complex is analogous to the present Panamanian Isthmus. The oceanic plate which lay to the east of this arc/trench boundary during the Early Mesozoic was necessarily consumed in subduction zones during rotation of the island arc. Thus the present Caribbean Plate, which once lay to the west of the Early Mesozoic arc/trench boundary, was derived from the Pacific area [Edgar et al., 1971; Malfait and Dinkelman, 1972] and emplaced eastward between North and South America.

Recent anomalous paleomagnetic data from the Cretaceous sediments of the Honduras platform [Gose and Schwartz, 1977] indicate that megatectonic rotation of small Central American crustal blocks is involved in the evolution of the Caribbean region. These detailed data, however, require both clockwise and counterclockwise rotations of the Honduras platform during the Cretaceous; this pattern of tectonic rotation is more complex than that proposed herein for the Greater Antilles and portions of northern Venezuela. It may simply reflect the different histories of rotation for the arc/trench complex of the Greater Antilles and northern Venezuela and the geologically distinct Honduras platform. Alternatively, the sparse time constraints on the paleomagnetic data from

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northern Venezuela may obscure a more complex pattern of tectonic rotation that is documented in the reasonably complete stratigraphic section of the Honduras. Future work may require a more complex interpretation than the relatively simple one adopted here.

Summary

Paleomagnetic data from plutonic rocks of Cretaceous age in the Caribbean Mountains of Venezuela are characterized by anomalous easterly declinations that cannot be satisfactorily explained by an anomalous magnetic field associated with a Cretaceous polarity transition interval. These anomalous data result from tectonic rotation of crustal units after acquisition of the NRM.

Most previous interpretations of the Caribbean Mountains have assumed a single east-west orogen that developed parallel to and along the present continental margin. Although not inconsistent with this model, the anomalous paleomagnetic declinations suggest also an alternative possibility that portions of the Caribbean Mountains were derived from a plate boundary perpendicular to the present continental margin and have since rotated and collided with the South American continent. Present data are insufficient to prove or disprove either model.

A compilation of paleomagnetic data from the entire Caribbean area indicates a pattern of predominantly westerly declinations along the northern Caribbean boundary and predominantly easterly declinations along the southern Caribbean boundary. If these magnetizations are contemporaneous, this pattern suggests that both the Greater Antilles and portions of the Caribbean Mountains may have formed a single north-south trending arc/trench complex which subsequently underwent megatectonic rotations during the Late Cretaceous.

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References

Alberding, H., Application of principles of wrench fault tectonics of Moody and Hill to northern South America, Geol. Soc. Amer. Bull., 68, 785-790, 1957.

Alvarez, W., Fragmented Andean Belt of Northern Colombia, Caribbean Geophysical, Tectonic and Petrologic Studies, Geol. Soc. Amer. Mem., 130, 77-96, 1971.

Beck, M. E., Jr., Discordant paleomagnetic pole positions as evidence of regional shear in the Western Cordillera of North America, Amer. J. Sci., 276, 694-712, 1976.

Bell, J. S., Tectonic evolution of the central part of the Venezuelan Coast Ranges, Caribbean Geophysical, Tectonic and Petrologic Studies, Geol. Soc. Amer. Mem., 130, 107-108, 1971.

Bell, J. S., Geotectonic evolution of the southern Caribbean area, Studies in Earth and Space Sicence, Geol. Soc. Amer. Mem., 132, 367-386, 1972.

Bellizzia, A. G., Sistema Montanoso del Caribe, Borde sur de la Placa Caribe es una Cordillera Aloctona?, Trans. Carib. Geol. Conf. VI, 247-258, 1972a.

Bellizzia, A. G., Is the entire Caribbean mountain belt of Northern Venezuela allochthonous?, Studies in Earth and Space Science, Geol. Soc. Amer. Mem., 132, 363-368, 1972b.

Bellizzia, A., M. Graterol, and D. Rodriguez, Ofiolitas de Siquisique y Rio Tocuyo y sus Relaciones con la Falla de Oca, Trans. Carib. Geol. Conf. VI, 182-183, 1972.

Case, J. E., L. G. Duran, A. Lopez, and W. R. Moore, 1971, Tectonic investigations in Western Colombia and Eastern Panama, Geol. Soc. Amer. Bull., 82, 2685-2712.

Compania Shell de Venezuela, Igneous rocks of the Siquisique Region, State of Lara, Bol. Inform. Asoc. Venez. Geol. Min. Petrol., 8, 286-306, 1965.

Coronel, G., Igneous rocks of Central Falcon, Bol. Inform. Asoc. Venez. Geol. Min. Petrol., 13, 155-162, 1970.

Creer, K. M., A paleomagnetic survey of South American rock formations, Phil. Trans. Roy. Soc. London, Ser. A, 267 (1183), 457-557, 1970.

Creer, K. M., and M. Sanver, The use of the sun compass, in Methods in Palaeomagnetism, edited by D. W. Collinson, K. M. Creer, and S. Runcorn, pp. 11-15, Elsevier, New York, 1967.

Dasgupta, S. N., and S. A. Vincenz, Paleomagnetism of a Paleocene pluton on Jamaica, Earth Planet. Sci. Lett., 25, 49-56, 1975.

Doell, R. R., and A. Cox, Palaeomagnetic sampling with a portable coring drill, in Methods of Palaeomagnetism, edited by D. W. Collinson, K. M. Creer, and S. K. Runcorn, pp. 21-25, Elsevier, New York, 1967.

Edgar, N. T., J. I. Ewing, and J. Hennon, Seismic refraction and reflection in the Caribbean Sea, Amer. Ass. Petrol. Geol. Bull., 55, 833-870, 1971.

Fink, L. K., and C. G. A. Harrison, Paleomagnetic investigations of selected lava units on Puerto Rico (abstract), Trans. Carib. Geol. Conf. VI, 5, 1971.

Goldstein, M., and D. Strangway, Paleomagnetism of a Miocene transition in southeastern Oregon, Earth Planet. Sci. Lett., 7, 231-239, 1969.

Gose, W. A., and D. K. Schwartz, Paleomagnetic results from Cretaceous sediments in Honduras: Tectonic implications, Geology, 5, 505-508, 1977.

Guja, N. J., and S. A. Vincenz, Equatorial paleopoles of some Cretaceous rocks on Jamaica and polarity transition periods (abstract), Eos Trans. AGU, 52, 188, 1971.

Harvey, S. R. M., Origin of the southern Carribbean Mountains, Studies in Earth and

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Space Sciences, Geol. Soc. Amer. Mem., 132, 387-400, 1971.
Jakes, P., and J. Gill, Rare earth elements and the Island Arc tholeiite series, Earth Planet. Sci. Lett., 9, 17-28, 1970.
La Rochelle, A., Adaptation of an astrocompass for collection of oriented rock specimens, Top. Rep., 32, 4 pp., Geol. Surv. of Can., Ottawa, 1961.
MacDonald, W. D., Estratigrafia, estructura y metamorfismo de las rocas del Jurasico Superior, Peninsula de Paraguana, Dir. Geol. Bol. Geol., 9, 441-456, 1968.
MacDonald, W. D., and N. D. Opdyke, Tectonic rotations suggested by paleomagnetic results from northern Colombia, South America, J. Geophys. Res., 77, 5720-5730, 1972.
MacDonald, W. D., B. L. Doolan, and V. G. Cordani, Cretaceous-Early Tertiary metamorphic K-Ar age values from the southern Caribbean, Geol. Soc. Amer. Bull., 82, 1381-1388, 1971.
Malfait, B. T., and M. G. Dinkelman, Circum-Caribbean tectonic and igneous activity and the evolution of the Caribbean Plate, Geol. Soc. Amer. Bull., 83, 251-272, 1972.
Maresch, W. V., Plate tectonics origin of the Caribbean Mountain system of northern South America: Discussion and proposal, Geol. Soc. Amer. Bull., 85, 669-682, 1974.
Martin Bellizzia, C., Edades isotopicas de rocas Venezolanas, Dir. Geol. Bol. Geol., 10, 356-380, 1968.
Martin Bellizzia, C., and J. de Arozena, Complejo ultramafico zonado de Tausabana-El Rodeo, Gabro zonado de Siraba-Capuana y Complejo subvolcanico estratificado de Santa Ana, Paraguana, Estado Falcon, Trans. Carib. Geol. Conf. VI, 337-356, 1972.
Menendez, A., Geology of the Tinaco area, north central Cojedes, Venezuela, Ph.D. dissertation, 238 pp., Princeton Univ., Princeton, N. J., 1962.
Menendez, A., Tectonics of the central part of the western Caribbean mountains, in Proceedings of the International Conference on Tropical Oceanography, pp. 103-130, University of Miami, Miami, Fla., 1967.
McElhinny, M. W., Paleomagnetism and Plate Tectonics, 358 pp., Cambridge University Press, New York, 1973.
Muessig, K. W., The Central Falcon Igneous Suite, Venezuela, Geol. Mijnbouw, 51 (2), 261-266, 1978.
Murray, C. G., Significance of possible Cretaceous Island Arc volcanism in the Venezuelan coast range (abstract), Trans. Carib. Geol. Conf. VI, 303, 1972a.
Murray, C. G., Petrologic studies of zoned ultramafic complexes of the Alaskan type: Feeder pipes of andesitic volcanoes, Studies in Earth and Space Science, Geol. Soc. Amer. Mem., 132, 313-335, 1972b.
North, F. K., The curvature of the Antilles, Geol. Mijnbouw, 44 (3), 73-86, 1965.
Oxburgh, E. R., Geology and metamorphism of Cretaceous rocks in eastern Carabobo state, Venezuelan Coast Ranges, Caribbean Geological Investigations, Geol. Soc. Amer. Mem., 98, 241-310, 1966.
Piburn, M. D., Metamorphism and structure of the Villa de Cura Group, northern Venezuela, Ph.D. dissertation, 148 pp., Princeton Univ., Princeton, N. J., 1967.
Rod, E., Strike-slip faults of northern Venezuela, Amer. Ass. Pet. Geol. Bull., 40, 457-476, 1956.
Roper, P. J., Plate tectonics, a plastic as opposed to a rigid body model, Geology, 2, 247-258., 1974.
Santamaria, F., and C. Schubert, Geochemistry and geochronology of the southern Caribbean - northern Venezuela plate boundary, Geol. Soc. Amer. Bull., 7, 1085-1098, 1974.
Seiders, V. M., Geology of central Miranda, Venezuela, Ph.D. dissertation, 255 pp., Princeton Univ., Princeton, N. J., 1962.
Shagam, R., Geology of central Aragua, Venezuela, Geol. Soc. Amer. Bull., 71, 235-362, 1960.
Skerlec, G. M., The western termination of the Caribbean Mountains of Venezuela: A progress report (abstract), Trans. Carib. Geol. Conf. VII, 493, 1974.
Skerlec, G. M., Geology of the Acarigua Region, western Venezuela, Ph.D. dissertation, 220 pp., Princeton Univ., Princeton, N. J., 1979.
Steinhauser, P., and S. A. Vincenz, Equatorial paleopoles and behavior of the dipole field during polarity transitions, Earth Planet. Sci. Lett., 19, 113-119, 1973a.
Steinhauser, P., and S. A. Vincenz, Paleomagnetism of Lower Cretaceous lavas on Jamaica, Arch. Meteorol. Geophys. Bioklimatol., Ser. A, 22, 325-336, 1973b.
Suppe, J., Interrelationships of high-pressure metamorphism, deformation and sedimentation in Franciscan tectonics, U.S.A., Tectonics-Tectonique, sect. 3, Proc. Int. Geol. Congr. 24th, 552-559, 1972.
Vincenz, S. A., Rock magnetism and paleomagnetic studies, Upper Mantle Project, final report, p. 142, Nat. Acad. of Sci. and Nat. Res. Counc., Washington, D. C., 1971.
Vincenz, S. A., and S. N. Dasgupta, Paleomagnetic study of some Cretaceous and Tertiary rocks on Hispaniola, Pure Appl. Geophys., 116, 1200-1210, 1978.
Vincenz, S. A., P. Steinhauser, and S. N. Dasgupta, Paleomagnetism of Upper Cretaceous ignimbrites on Jamaica, Z. Geophys., 39, 727-737, 1973.
Watkins, N. D., and F. W. Cambray, Paleomagnetism of Cretaceous dikes from Jamaica, Geophys. J. Roy. Astron. Soc., 22, 163-179, 1970.

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